

Objective: Let E_1 and E_2 be two sample images from some unknown distribution P_1 and P_2 , respectively. We want to test the following hypothesis

$$H_0 : P_1 = P_2 \quad \text{against} \quad H_1 : P_1 \neq P_2.$$

Vietoris–Rips (VR) complex: $VR_\epsilon = \{\sigma \subset X \mid \mathbb{D}(x_i, x_j) \leq \epsilon, \forall x_i, x_j \in \sigma\}.$

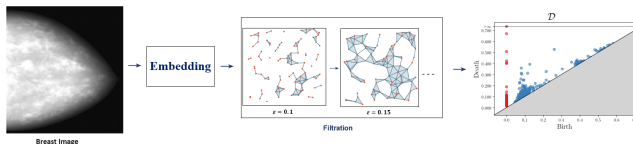


Figure: Illustration of the TDA on medical image.

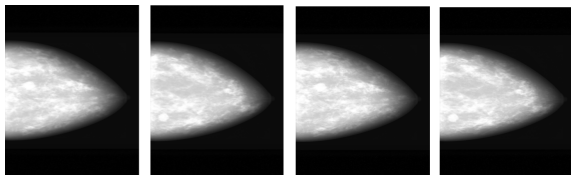
Persistence diagram: $\mathcal{D} = \left\{ \{(\epsilon_i^b, \epsilon_j^d) \in \mathbb{R}^2 \mid \epsilon_i^b < \epsilon_j^d\} \right\}_{\ell=1}^{\ell=n}$. From $\mathcal{D}$ we compute the **survival time** of each topological feature in the filtration process as $s_\ell = \epsilon_{\ell i}^b - \epsilon_{\ell j}^d$, and compute Kaplan–Meier (KM) estimate of the survival functions, $S(t) = P_r(T \geq t)$ as

$$\hat{S}(t) = \prod_{j: t_j < t} \left(1 - \frac{d_j}{n_j}\right).$$

Suppose $\hat{S}_x(t)$ and $\hat{S}_y(t)$ are the estimated survival functions (KM estimates) of the images E_1 and E_2 , respectively.

- **Test statistic:** $T_n = W(\hat{S}_x(t), \hat{S}_y(t))$, for some distance function W .
- We evaluate the critical value and p -value of the T_n based on Chi-squared distribution similar to *Log-Rank test*.

Comparing breast images for different relative radiation doses:



(a) Dose 20%, Case-1. (b) Dose 20%, Case-2. (c) 40%, Case-1. (d) Dose 40%, Case-2.

Hypothesis (p -value)	KS test	PD test	PL test	TDA-Survival test
Fatty, Relative Dose 20% versus Fatty, Relative Dose 40%	<0.001	<0.001	<0.001	<0.001
Fatty, Relative Dose 40% case 1 versus Fatty, Relative Dose 40% case 2	<0.001	<0.001	<0.001	0.206